

Lab 1: Requirements Engineering & UML Use-Case Modelling

Software Engineering

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Section : D

SRN: PES1UG25CS814

Disaster Relief Supply & Volunteer Coordinator:

Function and Non-functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall match incoming shelter supply requisitions with available central warehouse supplies and allocate delivery priority.	High	When a critical supply request is received, the system assigns it the highest dispatch priority and includes it in the logistics manifest.	Ensures critical supplies are dispatched first.	Directly based on the given FR.
FR-002	Functional	The system shall allow the Disaster Manager to create, update, and prioritize shelter supply requests for water, blankets, medical kits, and other essential supplies.	High	When the Disaster Manager creates or updates a request, the system stores the request and displays its assigned priority.	Supports effective management of shelter needs.	Should support different types of relief supplies.
FR-003	Functional	The system shall allow the Disaster Manager to assign Relief Volunteers	High	When a volunteer is assigned, the system verifies that the volunteer has	Ensures suitable volunteers are assigned.	Skill matching is important during emergencies.

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		based on their skills and availability.		the required skill and is available.		
FR-004	Functional	The system shall allow users to track relief vehicle deliveries, including delivery status and destination.	High	When a vehicle status is updated, the new status and destination are displayed correctly in the system.	Provides visibility of relief deliveries.	Status may include assigned, in transit, and delivered.
FR-005	Functional	The system shall allow users to report and record issues encountered during relief supply delivery.	Medium	When a delivery issue is reported, the system records the issue details and associates them with the relevant delivery.	Helps manage delivery problems.	Issue details should be recorded for follow-up.
NFR-001	Non-functional	The disaster management dashboard shall operate efficiently over low-bandwidth networks and support offline caching of essential GIS data.	High	Performance testing under simulated low-bandwidth conditions meets the defined latency target, and cached GIS data remains accessible when offline.	Ensures usability during poor network connectivity.	Based on the given NFR guideline.
NFR-002	Non-Functional	The system shall authenticate users and restrict access to disaster, volunteer, supply, and delivery information according to their assigned roles.	High	An unauthenticated user is denied access, and users can access only functions permitted for their assigned role.	Protects sensitive disaster-management information.	Includes authentication and role-based access control.

Security Requirements :

Security ID	Security Requirement	Priority	Acceptance Criteria
SR-001	The system shall require users to authenticate using a valid username and password before accessing the system.	High	An invalid username or password must prevent login and display an authentication error.
SR-002	The system shall provide role-based access control for Disaster Managers, Relief Volunteers, and Logistics Coordinators.	High	Each user can access only the functions permitted for their assigned role.
SR-003	The system shall prevent unauthorized users from viewing or modifying supply, volunteer, shelter, and delivery information.	High	An unauthorized access attempt is denied and the protected data remains unchanged.
SR-004	The system shall protect user and disaster-management data during transmission between the client and server.	High	Data transmitted between the client and server must use a secure communication mechanism.
SR-005	The system shall maintain an audit log of important actions such as supply allocation, volunteer assignment, delivery updates, and issue reporting.	Medium	Each important action records the user, action, and date/time.
SR-006	The system shall securely manage user sessions and automatically terminate inactive sessions after a defined timeout period.	Medium	An inactive session is automatically logged out after the configured timeout.